Tom Stepp

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Experience

Senior Software Engineer, Google Cloud Dataflow 2021 – Present

- Led data-plane design for Dataflow Streaming High Availability: zero data loss, exactly-once processing
- Designed persistent streaming shuffle RPCs: eliminated worker OOMs and cut regional fleet peak memory 28%
- Optimized Pub/Sub sink and asymmetric autoscaling, cutting customer pipeline costs by up to 70%
- Designed delegated Kafka deduplication: 55% lower read cost and 5x better tail latency
- Landed sub-sharded state commits on 13,000+ pipelines, cutting delivery latency by up to 50%

Software Engineer, Boeing AvionX 2017 – 2021

- Architect of continuous integration suite for flight software team of 150 developers
- Reduced Matlab code generation times from 10 to 1.5 hours with parallelization and caching

Software Engineer, Facebook 2016

- Created scripts for automation of network switch testing with Tcl and Ixia hardware API
- Developed driver for IC chip with C++ to improve Wedge100 switch function and reliability

Software Engineer, GE Aviation 2015

- Developed test validation software for Leap and Passport aircraft engines

Skills

C++, Java, Python, GCP Dataflow, Pub/Sub, GCS, Apache Beam, Kafka, Iceberg, distributed systems, stream processing.

Education

M.S. in Computer Science, **University of Southern California** 2021
B.S. in Electrical Engineering, **Purdue University** 2017
Study Abroad, **Universidad Carlos III de Madrid** 2016

Recognition

Apache Beam Summit Speaker: Four talks on Dataflow Streaming performance and scalability 2023 – 2026
Apache Beam College Instructor: Streaming Kafka Pipelines and Scaling Iceberg Ingestion 2025 – 2026
Google Perfy Awards: 1 Gold, 3 Silver, 2 Bronze for Dataflow Streaming performance work 2023 – 2026
Google Healthy Awards: Gold and Silver for code health and build infrastructure migrations 2025 – 2026
U.S. Patents: 3 patents granted for vehicle systems, filed at Boeing 2021 – 2026
Boeing Technical Journal: Analysis for Large-scale Software Systems 2021
Boeing Technical Excellence Conference: Three presentations on Model-Based DevOps 2020 – 2021

Leadership

Mentor & Technical Interviewer, Google 2022 – Present
STEM Outreach Volunteer, Boeing 2018 – 2021
ECE Student Society Board Member, Purdue 2014 – 2017
Eagle Scout, Boy Scouts of America 2013

Projects

ML Handwritten Digit Classification 2020

- Programmed neural network from scratch in Python to classify digits from MNIST database

Go-Playing AI Agent 2020

- Programmed Minimax algorithm in Python to beat other AI agents at games of 5x5 Go in real-time

Solr Search Engine 2021

- Created inverted index of news web pages, implemented auto-complete and spelling correction

Weenix Kernel 2021

- Implemented processes and threads, virtual filesystem, and virtual memory